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Regular Articles

Transfer learning with simulated variants and calculated quantities for nanobody melting-temperature prediction

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Thermal stability is important for nanobody production, storage, and formulation, but measuring melting temperature (Tm) for many sequences is costly. Computer simulations can produce related stability values, but they do not measure Tm itself. We tested whether those values improve Tm prediction with only 57 measured values for training. Models shared an ESM2 sequence encoder between Tm and one calculated quantity. Each was tuned with 114 validation Tm values and evaluated on the same 396 held-out values. We compared two ways of collecting molecular dynamics (MD) native-contact data. With a frozen encoder, local mutation scans of two fixed structures reduced mean absolute error by 0.20°C relative to Tm-only. A separately tuned heterogeneous-panel study reduced it by 0.05°C. The studies also differed in contact definitions, trajectory averaging, and model selection, so this difference cannot be assigned to variant choice alone. We also compared MD values with free-energy perturbation (FEP), Rosetta, and ThermoMPNN values. FEP gave the lowest observed error with both frozen and fine-tuned encoders. In the fine-tuned model, mean absolute error fell from 6.55°C to 6.40°C, although the 95% interval for the change included zero. MD native-contact values clearly helped only with the frozen encoder; Rosetta and ThermoMPNN values gave little or no improvement. More calculated values were not automatically more useful. The observed benefit differed with both the simulated variants and the quantity calculated.

Key words: nanobody thermostability, melting-temperature prediction, multi-task transfer learning, protein language model, molecular simulation labels

◄ ***Significance*** ►

Nanobodies need sufficient thermal stability for production, storage, and use. Melting temperature is a common measure, but testing many sequences is costly. Simulations can provide related stability data, but they do not measure melting temperature. A larger simulated data set does not necessarily give a better predictor. The best results in this study came from local mutation scans and free-energy calculations; several other calculated data sets gave little or no improvement. What to simulate and what to calculate should be decided with the intended experiment in mind.

Introduction

Machine-learning models for molecular design need measured examples, but many experiments are slow or expensive. Properties calculated with quantum chemistry, molecular simulation, or other physical models can be used as additional

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training data. This approach is often called simulation-to-real (Sim2Real) learning. It has been used in materials science to combine large computed data sets with smaller experimental data sets [1–3]. In some cases, prediction improves as more computed data are added [3]. This outcome is not guaranteed. A calculation may use an imperfect model, sample the wrong systems, or measure a property that is only weakly related to the experiment.

Protein stability makes this problem clear because several common labels have different physical meanings. Melting temperature (Tm) is the experimental temperature at which folded and unfolded populations are balanced under a given assay. A mutation free-energy change ($\Delta\Delta G$) instead describes the change in folding free energy caused by a sequence change. Rosetta and learned stability models provide scores on their own scales, while molecular dynamics (MD) can provide structural measurements such as the fraction of native contacts that remain during a trajectory. All of these labels concern protein stability, but none is a Tm measurement. A poorly matched computed label may therefore leave Tm prediction unchanged or make it worse.

We study this question in nanobodies, single-domain VHH antibodies used as small and modular binding proteins [4,5]. Thermal stability affects their expression, storage, and formulation [6–8]. Direct measurements and detailed simulations of stability remain costly [9–12]. Protein language models such as ESM2 can learn useful sequence features from large protein databases [13–15], and several methods use these features to predict Tm [16–21]. However, a sequence model does not remove the need for measured Tm values from the protein family of interest.

Large protein-stability data sets are becoming available. For example, Tsuboyama et al. measured folding stability for about 776,000 natural and designed protein domains by cDNA-display proteolysis [22]. Those domains are 40–72 residues long and differ from VHH domains in both sequence and measurement type. Models trained on small domains also transfer less well to longer proteins [23]. Nanobody-specific data and validation are therefore still needed [24–26].

The open question is not simply whether computed labels can be added to a Tm model. It is how to plan the simulations that produce those labels. A mixed set of unrelated sequences may contain unwanted differences such as sequence length. A mutation scan on one or two fixed structures gives a more controlled set of sequence changes. The calculated quantity also matters: a mutation free energy, a Rosetta score, and an MD native-contact value may differ in how closely they relate to Tm. Both the systems to calculate and the reported quantity are chosen before model training.

Here, we compare several computed labels in the same low-data Tm task. The labels are mutation free energies from free-energy perturbation (FEP), Rosetta mutation scores, ThermoMPNN scores, Rosetta scores for ESM2-proposed or random variants, and MD native-contact values. For the MD label, we compare two complete simulation plans: a heterogeneous structure panel and local mutation scans on the same 1MEL and 4IDL structures used for FEP. Both use 400 K MD and native-contact labels, but they also differ in contact selection and trajectory averaging. Their comparison therefore does not isolate the choice of variants. Each computed label is trained as a second task that shares an ESM2 sequence model with Tm prediction (Fig. 1). We tune every label separately using only the experimental Tm validation set. We then compare the selected models using paired errors on the same 396 test Tm labels, which were not used for training or model selection. In this paper, "transfer" means that adding a computed label lowers the test Tm error.

We observed a larger frozen-encoder gain under the local mutation-scan MD plan than under the heterogeneous-panel plan. Among the calculated quantities, FEP gave the lowest observed error in both encoder settings, whereas native-contact values gave a clear gain only with the frozen encoder. These results suggest that simulations intended to support an experimental predictor should be planned around both the systems to calculate and the quantity to report.

Materials and methods

Study design

We asked whether computed protein-stability labels could improve nanobody melting-temperature (Tm) prediction when only a small experimental training set was available. Experimental Tm validation error, never computed-label error, was used to select checkpoints and hyperparameters. Each computed label was added as a second regression task that shared all or part of the sequence model with the Tm task. We compared each label with a Tm-only model trained and selected in the same encoder setting.

Experimental Tm data

We used the thermo-tm data set from NbBench, a public nanobody benchmark [25]. We reassigned its three published partitions before the comparisons in this study so that the smallest partition was used for training. The resulting split contained 57 training, 114 validation, and 396 test sequences. This split models the setting that motivated the study: few Tm measurements for fitting a model, but enough test proteins for paired comparisons. The training set fit model parameters, the validation set selected checkpoints and hyperparameters, and the test set was kept for final evaluation.

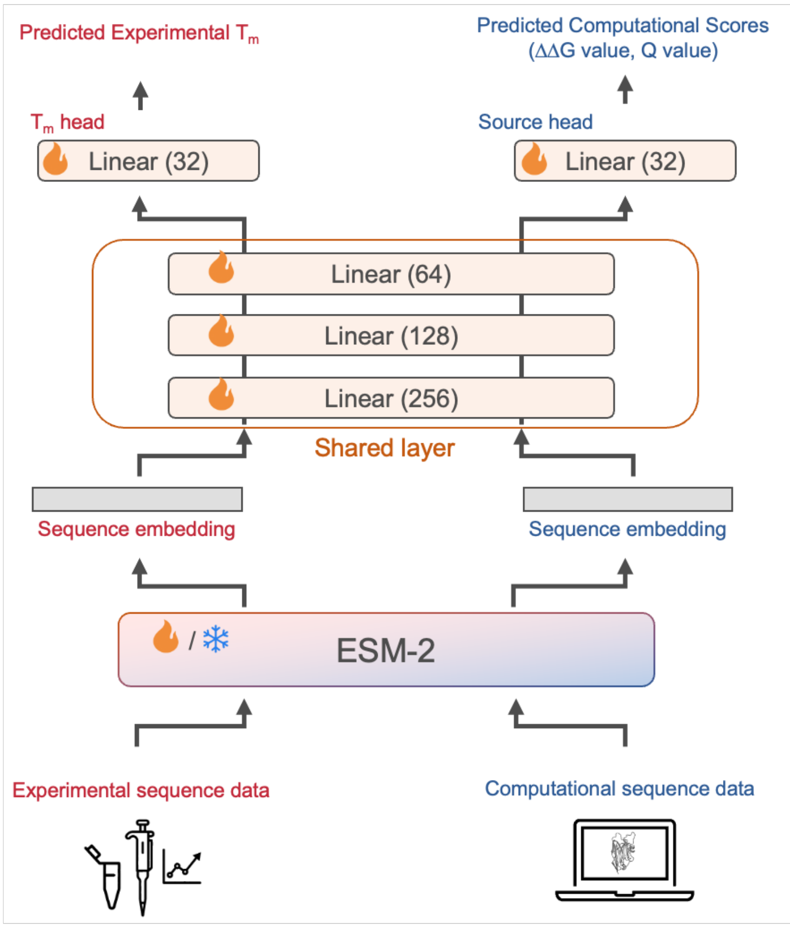

Figure 1 Multi-task model used to test computed stability labels. Experimental sequences and sequences with computed labels pass through the same ESM2 model and shared multilayer perceptron. Separate scalar heads predict Tm and the computed label. The ESM2 model is either kept frozen or fine-tuned. Models are selected using experimental Tm validation data, and the final comparison uses held-out Tm test data.

Tm values were stored in degrees Celsius. We fit a robust scaler and then a min–max scaler to [0,1] using only the 57 training values. The fitted scalers were applied to the training and validation labels. Predictions were returned to degrees Celsius before validation and test errors were calculated. Validation and test Tm values therefore did not affect the scaling step.

Computed-label tables and sampling

We tested seven sources of computed labels: FEP mutation free energies, matched MD native-contact values, Rosetta mutation scores, ThermoMPNN predictions, Rosetta scores for random variants, Rosetta scores for ESM2-proposed variants, and MD native-contact values for a heterogeneous structure panel. These values were treated as related measurements, not as estimates of Tm.

FEP, Rosetta, and ThermoMPNN used two single-mutation tables based on the 1MEL and 4IDL structures. Each contained 435 and 409 rows, respectively. The matched MD tables used the same two structures and mutation scans and contained 431 1MEL rows and 406 4IDL rows that reached the common analysis time. All 837 matched-MD sequences were present in the 844-row FEP pool; seven FEP variants did not have a matched MD label. The random and ESM2-proposed variant sets each contained 1000 two-mutation sequences per structure. The heterogeneous MD table contained 1143 PDB-derived rows representing 833 unique amino-acid sequences. The sampling unit for this table was a PDB row, so a sequence represented by more than one structure could occur more than once.

For all two-table sources, the notation *n* means a cap of *n* rows from *each* structure table for each ensemble member, not

n rows in total. Thus, at $n = 320$, each member sampled at most 640 computed-label rows. The heterogeneous MD source had one table, so n is the total number sampled for each member. Each run used its own seed to sample rows. Sampling was carried out separately for each label source; in particular, the FEP and matched-MD runs were not forced to select the same mutations from their nearly identical pools. After sampling, 80% of the rows were used for training and 20% were set aside. The set-aside rows were not used to choose checkpoints; only the experimental Tm validation rows were used for that purpose. The relation between sequence length and the heterogeneous-panel MD label is shown in Supplementary Fig. S1; exact sequence matches to the target data are reported below.

The 1MEL and 4IDL tables were kept as two tasks because their score ranges can differ. Models used either two separate output heads or one output head with a learned structure indicator, as described below. Every processed table supplied an amino-acid sequence and one scalar label scaled to [0,1]. FEP, Rosetta, and ThermoMPNN scores were sign-reversed so that larger values indicated a more favorable change, then processed separately for each structure with robust scaling, a Yeo–Johnson transform, and min–max scaling. MD contact values were min–max scaled with larger values indicating greater contact retention.

We also checked exact sequence matches between every computed-label table and the NbBench splits. There were no exact matches for the FEP, matched MD, Rosetta, ThermoMPNN, random-variant, or ESM2-proposed tables. The heterogeneous MD table contained 2 training, 4 validation, and 8 test sequences also present in NbBench. When one of these rows was sampled for auxiliary training, the model saw the target sequence with a computed Q label. It did not see the experimental validation or test Tm value, which remained unused in training. We retained the rows and report them because the heterogeneous-panel comparison was therefore not fully disjoint from the target sequences.

FEP mutation free energies

The FEP labels were mutation free-energy changes, $\Delta\Delta G$, for the 1MEL and 4IDL mutation scans. Positive raw $\Delta\Delta G$ denotes a destabilizing mutation.

The fixed FEP tables were produced with alchemical simulations in NAMD using the CHARMM36 protein force field and explicit TIP3P water [27–29]. The starting coordinates were the 1MEL and 4IDL crystal structures. Mutation systems were prepared with the VMD AlaScan workflow and CHARMM hybrid-residue topologies [30, 31]. Representative archived input files show a 2-fs timestep, particle mesh Ewald electrostatics, bonds to hydrogen held fixed, a temperature of 300 K, and a pressure of 1 atm [32, 33].

Each forward and backward transformation covered 20 windows of width $\Delta\lambda = 0.05$ between 0 and 1. A window contained 500,000 steps (1 ns), with its first 100,000 steps assigned to equilibration. The folded-state free energy was combined with a residue-specific unfolded-state reference to obtain $\Delta\Delta G$ relative to the wild type.

MD native-contact labels

Both MD labels were based on the smooth native-contact fraction Q [34]. For a native contact with reference distance d_{ij}^0 and distance $d_{ij}(t)$ at time t, its contribution was

$$q_{ij}(t) = \left[1 + \exp\left\{\beta[d_{ij}(t) - \lambda d_{ij}^0]\right\}\right]^{-1},$$

with $\beta = 50\ \mathrm{nm}^{-1}$ and $\lambda = 1.8$. Native contacts were pairs within 0.45 nm in the reference frame and separated by more than three residues. Q was the mean over the selected contacts and frames.

Matched mutation scans. We simulated the 1MEL and 4IDL wild types and the same single-mutant sequences used in the FEP comparison. Each protein was placed in a cubic box of OPC water with a 15-Å buffer and 0.15 M salt, and was described by Amber ff19SB with hydrogen-mass repartitioning [35, 36]. OpenMM was used for minimization, equilibration, and production [37]. The recorded run settings were 0.2 ns NVT followed by 0.8 ns NPT at 300 K, 0.5 ns restrained NVT at 400 K, and restraint-free NVT production at 400 K. All stages used a 4-fs timestep; production coordinates were written every 10 ps. Although some runs continued to 100 ns, the label used the first 4000 production frames (40 ns) for every variant.

For this source, the reference distances came from production frame 0 and all backbone atoms selected by MDAnalysis were eligible for contacts. We computed $\Delta Q = Q_{\mathrm{mutant}} - Q_{\mathrm{WT}}$ within each structure and then min–max scaled it. Subtracting one constant wild-type value does not change a min–max-scaled label, so scaled Q and scaled ΔQ are identical within each structure table.

Heterogeneous nanobody panel. The comparison panel was built from SAbDab structures [38]. One verified heavy-chain nanobody was taken per PDB entry when an IMGT-renumbered structure was available. Preparation used PDBFixer and pH-aware protonation with pdb2pqr/PROPKA when available [39, 40]. The systems used ff19SB, OPC water, 0.15 M salt, and OpenMM. After 300 K NVT and NPT equilibration, two restrained 400 K NVT stages preceded 100 ns of unrestrained 400 K NVT production. Requiring a usable topology and trajectory, a sequence of at least 50 residues, and at least one selected contact gave 1143 rows.

The heterogeneous-panel label used production frame 0 as its reference and averaged approximately the last 30 ns. Its contact set contained backbone heavy-atom pairs for which at least one residue was Asp, Glu, Gln, Asn, Arg, Lys, or His. It was min–max scaled across the single table. The matched and heterogeneous studies used trajectory lengths in the same broad range: tens of nanoseconds for the matched scans and 100 ns for the heterogeneous panel. They used the same Best–Hummer form and the same temperature, although their contact selection and averaging window were not identical. We therefore treat this as a comparison of complete simulation plans, rather than as a one-variable test of sequence choice alone.

Rosetta, ThermoMPNN, and variant-selection comparisons

Rosetta scores were calculated with the `ddg_monomer` application and the `ref2015` score function [41,42]. The archived command used 50 iterations, Cartesian minimization, minimum-score aggregation, and full-atom input. The same settings were used for the 1MEL/4IDL mutation tables and for the two sets of two-mutation variants.

For the random set, mutation positions and replacement amino acids were sampled uniformly, excluding cysteine as a replacement, until 1000 unique two-mutation variants had been obtained for each structure. For the ESM2-guided set, the 650M-parameter ESM2 model generated 100,000 two-mutation variants per structure [14]. Variants were ranked by pseudo-log-likelihood, and the top 1% were scored with Rosetta. ThermoMPNN predictions were made for the same 435 and 409 single-mutation sequences used by the structure-matched tables [43]. We used the two processed ThermoMPNN tables, not the larger raw Colab outputs stored with them.

Sequence model and regression heads

The main encoder was the 8M-parameter ESM2 model `facebook/esm2_t6_8M_UR50D` [14]. The tokenizer used a maximum length of 160 tokens, including the beginning and end tokens, so at most 158 amino acids were supplied to ESM2. Longer sequences were truncated at the end. All NbBench target sequences were 151 residues or shorter. The first-token representation was used as the sequence vector. In the shared model, this vector passed through linear layers of size 256, 128, and 32, with ReLU activation and dropout after the first two layers. The Tm head and each computed-label head were separate 32-to-1 linear layers.

The model search also included residual and latent forms. Both gave Tm and computed-label examples separate 256–128–32 paths. The residual form added a sigmoid-gated correction to a base Tm prediction. The latent form predicted a Tm correction from the Tm vector, the second path's vector, and their elementwise product. For two-table sources, we compared separate 1MEL and 4IDL heads with a single head conditioned on a learned structure embedding. In the residual and latent implementations, the computed-label loss received a detached encoder vector by default, whereas the Tm path could still update the encoder. In the shared form, losses from all tasks could update the encoder when it was not frozen.

We tested two encoder settings. In the frozen setting, all ESM2 parameters were fixed and sequence vectors were computed once before head training. In the fine-tuned setting, all ESM2 parameters could change. AdamW used a smaller learning rate for ESM2 than for the newly initialized layers. Selected controls used 35M- and 650M-parameter ESM2 encoders.

Loss and model selection

Each row had one task identifier and contributed loss only through that task's output head. We used Huber loss with $\delta = 1$ on the scaled labels. By default, the losses were combined with learned uncertainty weights [44]. For task k, loss L_k, and learned log-scale s_k, the term added to the total loss was

$$\frac{L_k}{2\exp(2s_k)} + s_k.$$

Fixed task weights were tested as controls.

Models were trained with AdamW, cosine learning-rate decay, a batch size of 16, 100 warmup steps, and at most 400 epochs. Mixed-precision training was used on CUDA devices. A checkpoint was saved and evaluated once per epoch. The checkpoint with the lowest mean squared error on the 114 experimental Tm validation rows was retained, and training stopped after 15 epochs without improvement.

For the structure-matched label comparison in Fig. 3, each label source was tuned separately in the frozen and fine-tuned settings. The search had two stages. Stage 1 crossed three model forms (shared, residual, and latent) with two head forms (separate and structure-conditioned); Tm-only models searched the three model forms without a computed-label head. Stage 2 held the chosen form fixed. For fine-tuned models it tested encoder learning rates of 1×10^{-4}, 5×10^{-5}, 3×10^{-5}, and 1×10^{-5}. For frozen models it tested head learning rates of 1×10^{-3}, 5×10^{-4}, and 2×10^{-4}. Both searches crossed dropout values of 0.05, 0.15, and 0.30 with weight decay values of 0.02 and 0.10.

Each candidate was ranked by the MAE of a three-seed ensemble on the experimental Tm validation set, using $n = 320$ rows per structure table. The selected setting was then trained with five seeds for the final test result. Label-count curves used the same selected setting at $n = 20$, 80, 160, and 320 per structure; they were not retuned at each point. The heterogeneous-panel study was selected separately with its own Tm-only reference. Its final ensembles used ten seeds and $n = 640$ rows from the single heterogeneous table, the same total number sampled per ensemble member as $n = 320$ from each of the two matched tables. The run seed controlled randomness in Python, NumPy, PyTorch, and CUDA, as well as computed-row sampling and the computed-label training–monitoring split.

Test evaluation and statistics

For each condition, the five matched-label models, or the ten heterogeneous-panel models, predicted all 396 test sequences. Their predictions were averaged in scaled space and then returned to degrees Celsius. The main measure was mean absolute error (MAE). We also calculated root mean squared error, R^2, and Spearman and Pearson correlations.

Uncertainty was estimated by nonparametric bootstrap over test proteins. For one condition, we resampled its 396 absolute errors with replacement 10,000 times, using random seed 42, and calculated MAE for each sample. The reported 95% confidence interval is the 2.5th to 97.5th percentile range.

For a comparison, the same resampled test indices were applied to the candidate and reference error vectors. We calculated ΔMAE as candidate MAE minus reference MAE for every sample. Negative values therefore favor the candidate. Main figures report the mean paired ΔMAE and its 95% confidence interval. We do not use a separate p-value threshold for the main claims.

Software and saved outputs

The data loader, training code, model settings, and ensemble evaluation are in `prepare.py`, `train.py`, and the scripts distributed with the repository. Each final run stores its command-line settings, predictions, absolute errors, summary measures, and bootstrap results in machine-readable files. The figure scripts read those saved files directly. The released procedure begins with the fixed processed CSV tables; it does not repeat the raw FEP, Rosetta, ThermoMPNN, or MD calculations.

Results and discussion

Computed labels were judged only by experimental Tm

We tested each computed label as a second regression task (Fig. 1). The computed values were not treated as extra Tm measurements. ESM2 was either frozen or fine-tuned, and only the Tm validation error was used to choose models and settings. We used 57 measured Tm values for training, 114 for validation, and 396 for final testing. The paired comparisons used the same Tm test set for every model.

The selected Tm-only model had a test MAE of 7.23°C with a frozen encoder and 6.55°C with a fine-tuned encoder. We used each baseline only for comparisons within the same encoder setting. We then asked two questions. First, how did the results differ between the two MD plans? Second, which computed labels gave the most consistent results when ESM2 was frozen or fine-tuned?

The matched mutation-scan plan gave a larger frozen-encoder gain

We compared two MD simulation plans. The heterogeneous plan sampled 1143 PDB structures representing 833 unique sequences of different lengths. The matched plan used local mutations of the fixed 1MEL and 4IDL structures. Both used 400 K MD, trajectory lengths of tens to 100 ns per sequence, and labels from the Best–Hummer native-contact Q family. Their contact definitions, averaging windows, and model searches were not identical. This is therefore a comparison of two complete plans, rather than a test in which only the selected sequences were changed.

The two studies were run and tuned separately, so we did not compare their absolute MAEs. Figure 2a instead shows the paired change from the Tm-only model in each study. In the heterogeneous study, the MD label changed MAE by −0.049°C with a frozen encoder (95% CI, −0.092 to −0.006°C) and by +0.116°C with a fine-tuned encoder (95% CI,

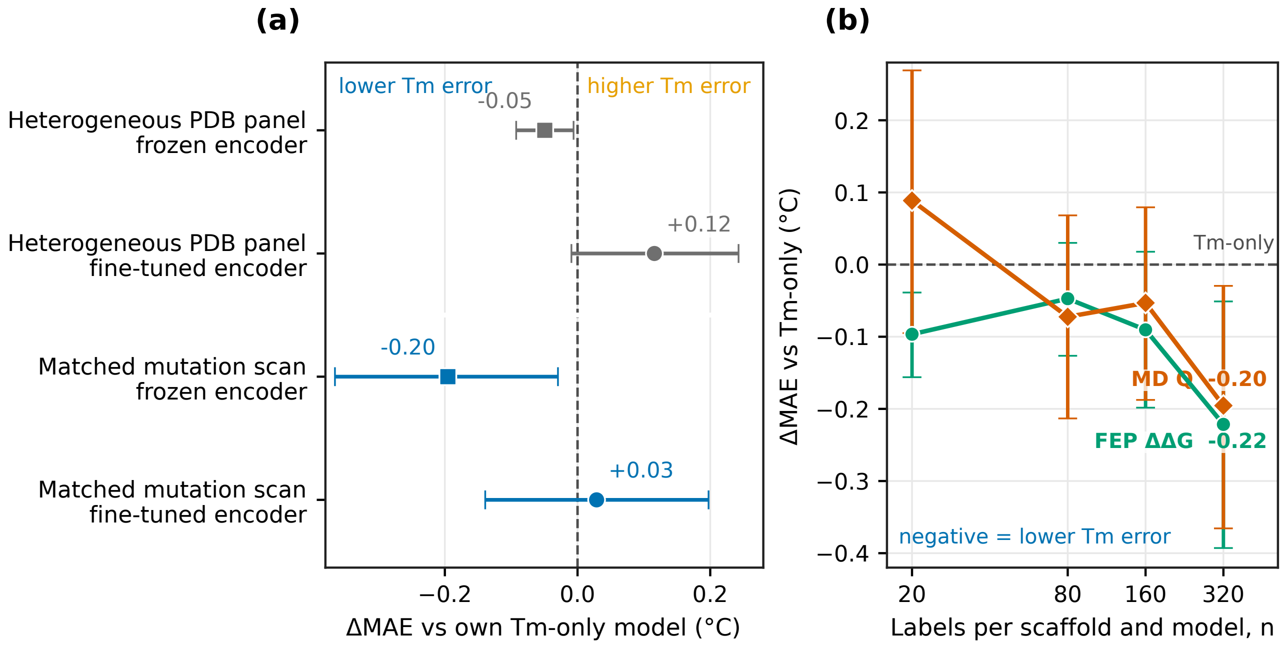

Figure 2 The matched mutation-scan plan showed a larger observed frozen-encoder gain from native-contact Q. The heterogeneous plan distributed simulations across PDB structures of different lengths, whereas the matched plan used local mutations of 1MEL and 4IDL. Both used 400 K MD and labels from the same Q family, but their contact definitions, averaging windows, model searches, and Tm-only references differed. (a) Paired MAE change for each plan and encoder setting. Zero means equal MAE to that study's own Tm-only model, and negative values mean lower Tm error. The heterogeneous and matched studies used fixed 10-seed and five-seed ensembles, respectively. (b) Paired MAE change from the frozen Tm-only model over four label counts. Here n is the number sampled from each structure table for each ensemble member; 80% of the sampled rows entered training. Points and whiskers show the paired change and its 95% bootstrap interval over the 396 Tm test values.

−0.009 to +0.243°C). In the matched study, the frozen change was −0.195°C (95% CI, −0.366 to −0.030°C), a larger observed gain than in the heterogeneous study. The fine-tuned change was +0.029°C (95% CI, −0.139 to +0.197°C).

The heterogeneous table contained eight sequences from the Tm test set. The model saw computed Q values, but not the test Tm values, for these sequences. When the eight sequences were omitted from the error calculation, the paired changes were almost unchanged: −0.049°C for the frozen encoder (95% CI, −0.094 to −0.005°C) and +0.122°C for the fine-tuned encoder (95% CI, −0.005 to +0.248°C). Thus, the test errors of these eight sequences did not drive the contrast in Fig. 2a. We did not retrain after removing the overlapping computed-label rows, however, so this check does not rule out an effect on training or model selection.

Several features of the two plans may have contributed. In the heterogeneous table, Q had a Pearson correlation of r = +0.13 with sequence length (Supplementary Fig. S1), so the model could partly use length instead of the intended contact information. The matched scans held length fixed within each structure and described local changes caused by one mutation. The heterogeneous table also sampled PDB rows, so repeated structures of one sequence received more weight. These observations do not show that any one feature caused the difference.

Subtracting a structure-specific wild-type constant does not affect later min–max scaling, so scaled Q and scaled ΔQ were identical within each matched table.

The frozen label-count curves provide a second view of the matched plan (Fig. 2b). At n = 20, the paired changes were −0.10°C for FEP and +0.09°C for matched MD. At n = 320, they were −0.22°C for FEP and −0.20°C for matched MD, and both 95% intervals were below zero. The direct FEP-minus-MD difference at n = 320 was −0.03°C (95% CI, −0.24 to +0.19°C), so we did not resolve a difference between the two labels with a frozen encoder. The four settings were not monotonic enough to support a scaling law.

FEP gave the lowest observed error in both encoder settings

We next compared all computed labels after each was tuned separately (Fig. 3a). With a frozen encoder, FEP and matched MD gave the clearest gains. FEP reached 7.01°C MAE, a paired change of −0.221°C from Tm-only (95% CI, −0.393 to −0.051°C). Matched MD reached 7.03°C, a change of −0.195°C (95% CI, −0.366 to −0.030°C). ThermoMPNN was third at 7.09°C, but its interval included zero. The Rosetta-based labels were close to or worse than the Tm-only model.

With a fine-tuned encoder, FEP reached 6.40°C, the lowest observed MAE in the study. Its paired change from the 6.55°C Tm-only model was −0.153°C (95% CI, −0.323 to +0.020°C). The interval included zero. Matched MD, ThermoMPNN, plain Rosetta, and random variants plus Rosetta differed from Tm-only by +0.029 to +0.144°C, with intervals that also included zero. ESM2-proposed variants plus Rosetta had higher error by +0.411°C (95% CI, +0.140 to +0.698°C).

Figure 3b directly compares FEP and matched MD. Zero means equal Tm test-set MAE, and negative values favor FEP. The frozen difference was −0.026°C (95% CI, −0.243 to +0.192°C). With a fine-tuned encoder, the observed difference was −0.182°C (95% CI, −0.393 to +0.024°C). Neither interval excluded zero at the largest label count.

The label-count curves show how these observed differences developed (Fig. 3c). At $n = 20$, matched MD had higher error than Tm-only by +0.47°C (95% CI, +0.26 to +0.69°C). Its error decreased as n increased, and its interval included zero by $n = 160$. FEP changed from +0.02°C at $n = 20$ to −0.15°C at $n = 320$, but the 95% interval at the largest setting also included zero. These four settings show recovery and crossover, not a power law.

FEP $\Delta\Delta G$ and Tm are different quantities, but both are linked to folding thermodynamics. Here, Tm supplied absolute temperatures across nanobodies, whereas FEP supplied local stabilizing and destabilizing changes in two structures. Native-contact retention is a more indirect measure of folding stability. This difference offers one possible explanation for the frozen and fine-tuned results, but the present comparison cannot establish the mechanism.

A recent preprint from our group examined ESM2 after fine-tuning on nanobody Tm with sparse autoencoders [45]. It found features linked to surface charge, hydrophobic-core packing, the VHH tetrad, and disulfide bonds. Together with that work, the present result suggests a testable question: do FEP labels support some of the same local stability features that appear during Tm fine-tuning? We did not examine hidden representations here, and the selected FEP and MD models did not always use the same regression form. Comparing the learned features of Tm-only, Tm+FEP, and Tm+MD models is therefore a separate next step.

Checks and limits

Exploratory fixed-setting tests with the 35M- and 650M-parameter ESM2 models showed the same direction as the selected 8M model: adding FEP gave a lower observed MAE than Tm-only within each model size (Supplementary Fig. S2a). The larger models were not searched in the same way as the 8M model, so these tests do not provide a model-size comparison. An analytical correction for periodic images in charge-changing FEP mutations had almost no effect on the scaled labels (Supplementary Fig. S2b).

ESM2-proposed variants scored by Rosetta did not have lower error than random variants scored by the same method. This result does not test a full protein design study; it shows that proposing a sequence and scoring it must be judged together.

Computed data should be judged by how well they help the measured task, and the choice of proteins and calculations should be made before spending the simulation time. A large mixed set is not automatically better than a smaller local scan. FEP and matched MD used roughly similar simulation time per variant here, but we did not save wall-clock costs in a form that supports a broader cost claim.

This study has several limits. The Tm training set contains only 57 nanobodies, and the matched labels cover two structures. The two MD plans differ in more than their sequence sets. The heterogeneous table sampled repeated PDB entries, contained eight exact matches to the Tm test sequences, and included 11 unusually long rows. The selected FEP and MD models also differed in regression form and in whether the auxiliary loss updated the encoder. The bootstrap intervals resample the fixed set of test proteins after averaging the trained models; they do not include uncertainty from choosing auxiliary rows, training seeds, or model settings. Finally, we did not make new Tm measurements or test predictions on newly measured nanobodies.

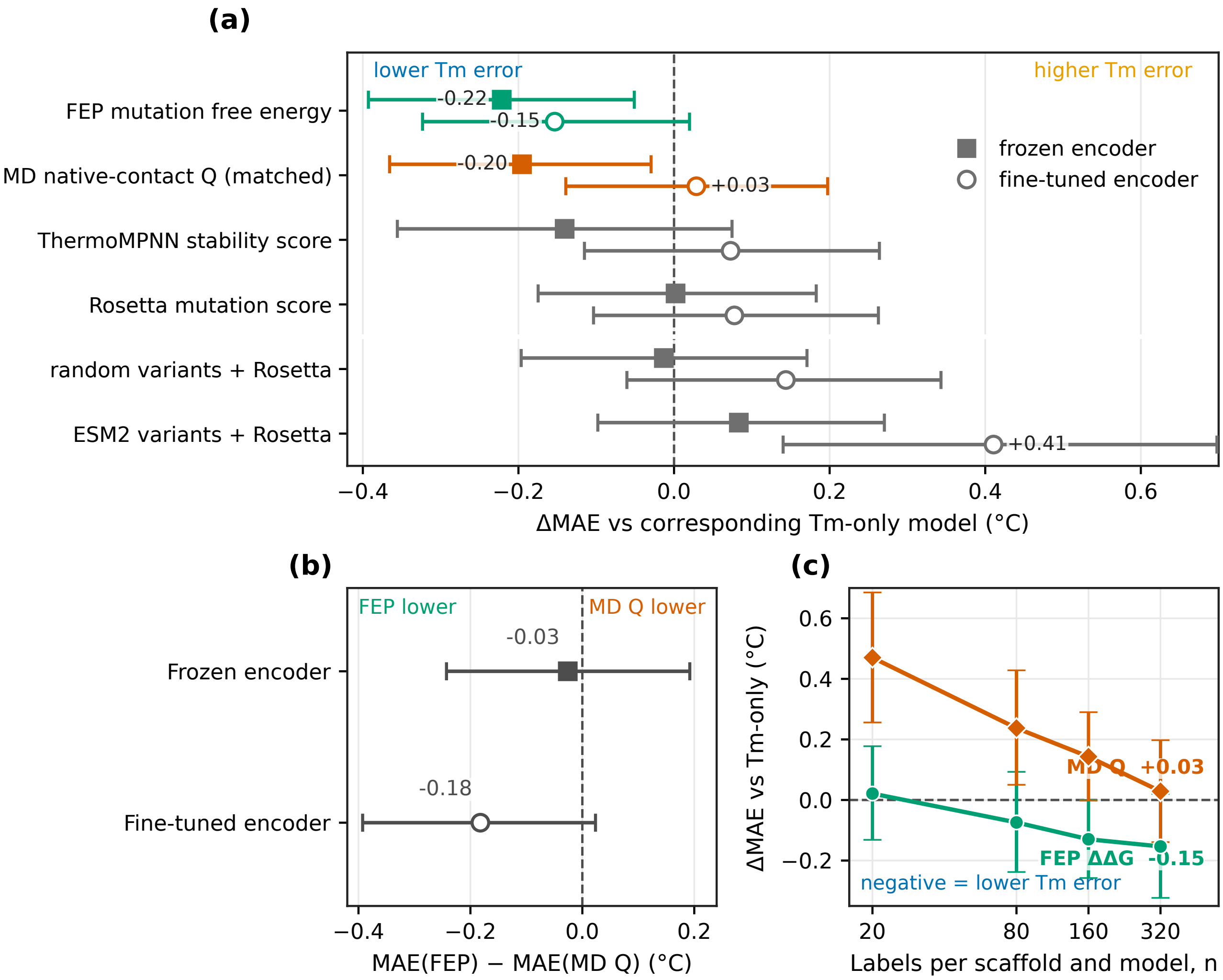

Figure 3 FEP gave the lowest observed Tm error in both encoder settings. (a) Paired MAE change from the corresponding Tm-only model for every tuned label. Negative values mean lower Tm error. Filled squares and open circles denote frozen and fine-tuned encoders, respectively. (b) Direct FEP-minus-MD comparison. Zero means equal Tm test-set MAE, negative values favor FEP, and positive values favor MD native-contact Q. (c) Paired MAE change from the fine-tuned Tm-only model over four label counts. Here n is the number sampled from each structure table for each ensemble member; 80% entered training. Whiskers in all panels are 95% bootstrap intervals over the 396 Tm test values.

Conclusion

Computed labels improved low-data nanobody Tm prediction only in specific settings. Among the complete MD plans tested here, native-contact labels from local mutation scans on 1MEL and 4IDL gave a larger observed frozen-encoder gain than labels from a heterogeneous structure panel. The plans differed in more than the choice of variants, so this result does not assign the difference to one factor. FEP mutation free energies gave the lowest observed error in both encoder settings, although the 95% interval for the fine-tuned comparison included zero. Matched-scan MD native-contact values gave a clear gain only with the frozen model. Together, these results suggest planning simulations around both the systems to calculate and the quantity to report, rather than around data volume alone.

In the longer term, matched scans could be extended to more structures and used in an iterative design study. A model could choose which variant to calculate next, receive an FEP or MD result, and use that result to choose the following variant. Reinforcement learning is one way to make these sequential choices, but the present results caution against using a computed score as the only reward. Our comparison of ESM2 and random proposals under the same Rosetta score shows why variant generation and scoring must be tested together. Any future reinforcement-learning study should therefore be compared with random sampling and simpler active-learning methods, and its success should be tested with new experimental Tm measurements.

Conflict of interest

T.M. is an employee of Epsilon Molecular Engineering, Inc. Y.M. is a scientific advisor of Epsilon Molecular Engineering, Inc.

Author contributions

Y.M. conceived and designed the study. Y.M., T.M., and K.S. carried out the overall computational study. Y.M. and T.M. analyzed the data. S.O. and K.O. performed the FEP calculations. Y.M. and T.M. wrote the manuscript. All authors discussed the results, reviewed the manuscript, and approved the final version.

Data availability

The source code for data processing, model training, analysis, and figure generation is available at `https://github.com/matsunagalab/sim2real`. The processed label tables, per-sequence test errors, and figure source data will be available from Zenodo (DOI: `10.5281/zenodo.XXXXXXX`). Raw MD and FEP files are not included because of their size and are available from the corresponding author upon reasonable request.

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